# DP Computer Science: A 1.2.5 Logic Diagrams

## 1. Core Concepts & Learning Objectives

### Learning Intentions

By the end of this lesson, students will be able to:

- **Construct** a logic diagram from a Boolean expression, using the correct standard symbol for each of the seven gates and labelling every input and the output
- **Trace** a set of inputs through a logic diagram to produce its output
- **Construct** the diagram for a combination of gates that performs a more complex operation, such as the **half adder**
- **Simplify** a logic diagram or expression by applying the laws of Boolean algebra, naming the law used at each step

> **Command Term Note:** The lesson is held at **Construct**—students actually draw circuits and simplify expressions. XNOR is now included (full set of seven gates). Boolean algebra laws and diagram construction are taught here, not in A1.2.4.

### Key Vocabulary (Tier 3)

| Term | Definition |
|---|---|
| **Logic diagram** | A drawing of a circuit as gates joined by wires, with signals flowing left to right |
| **Standard gate symbol** | The agreed shape for each gate: AND, OR, NOT, NAND, NOR, XOR, and XNOR |
| **Bubble** | The small circle on a gate's output that inverts it—turning AND into NAND, OR into NOR, and XOR into XNOR |
| **Boolean-algebra law** | A rule that rewrites an expression as an equivalent one, such as identity, complement, or absorption |
| **De Morgan's theorem** | `NOT(A · B) = Ā + B̄` and `NOT(A + B) = Ā · B̄`; a NOT over a bracket flips AND and OR and inverts each variable |
| **Half adder** | Two gates on the same two inputs—an XOR for the sum and an AND for the carry—which add two single bits |

### Pedagogical Focus

- **Self-Management Skills (ATL):** Students consciously practise self-management by organising a multi-step simplification—one named law per line, working from the inside of the brackets outwards—and by checking their own work by re-tabulating against the original.
- **Assessment (ATT):** This lesson runs on visible evidence. Share at the outset what a correct diagram and a fully simplified expression look like, then use mini-whiteboards and a hinge question partway through to see who has the gate symbols and who is confusing them.

**Name the skill to students:** "*Today you are managing and checking your own working—the truth table tells you whether you're right without waiting for me.*"

### EAL Scaffold

| Support Type | Description |
|---|---|
| **Gate Symbol Reference Strip** | Including XNOR, kept on the desk |

| Support Type | Description |
| --- | --- |
| **Fully Worked Simplification** | Each law named, kept as a model to imitate |
| **Sentence Frame** | "By the [law name] law, [expression] simplifies to [result] because ______." |

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## 2. Lesson Sequence (One-Hour Block)

### Part 1: Do Now (Retrieval) — ~10%

**Activity:** Three short recall items:

| Question | Source |
| --- | --- |
| 1. How do you confirm that two circuits are equivalent? | [A1.2.4] |
| 2. Which gate is 1 only when its inputs differ, and which only when they match? | [A1.2.3] |
| 3. NAND, NOR, and XNOR are which three gates with their output inverted? | [A1.2.3] |

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### Part 2: Main Sequence — ~60%

#### Phase 1: Success Criteria (~5 mins)

**What a Correct Diagram Needs:**

| Criteria | Description |
| --- | --- |
| **Correct symbol** | The agreed shape for each gate |
| **Correct connections** | Wires that fix the signal path |
| **Labels** | Every input and the output labelled |

**What "Fully Simplified" Means:**

- Each step is a **named law**
- Nothing reduces further
- The result can be **verified** by re-tabulation (truth table)

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#### Phase 2: The Seven Symbols and a Hinge Check (~10 mins)

**The Three Building Blocks:**

| Gate | Symbol | Rule |
| --- | --- | --- |
| AND | `&` | 1 only when both inputs are 1 |
| OR | `≥1` | 1 when either input is 1 (including both) |
| NOT | `¬` | Flips its single input |

**The Bubble Rule — Derived Gates:**

| Gate | Relationship | Rule |
| --- | --- | --- |
| **NAND** | AND + bubble | Inverted AND |
| **NOR** | OR + bubble | Inverted OR |
| **XNOR** | XOR + bubble | Inverted XOR |

**The Exclusive Pair:**

| Gate | Rule |
| --- | --- |
| **XOR** | 1 only when inputs differ |
| **XNOR** | 1 only when inputs match |

**Hinge Question (Whiteboards):**

1. "*Draw the XNOR symbol.*"
2. "*Which single change turns an OR into a NOR?*"
3. "*Which gate is 1 only when the inputs match?*"

> **Read the boards and reteach the confusable pairs before moving on.**

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**Phase 3: Construct from an Expression (~15 mins)**

**Worked Example — Constructing a Diagram:**

> *Construct the diagram for:*
> `Z = (A · B) + C̄`

**Step-by-Step Construction:**

| Step | Action |
| --- | --- |
| 1 | Work from **inside the brackets outwards** |
| 2 | Draw the AND gate for `A · B` |
| 3 | Draw the NOT gate for `C̄` |
| 4 | Draw the OR gate to combine them |
| 5 | Label every input and the output |

```text
A ──┐
    ├── AND ──┐
B ──┘         │
              ├── OR ── Z
C ─── NOT ────┘
```

**Student Task:** Construct a diagram from a given expression and **self-check** against the criteria.

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**The Half Adder — Combining Gates:**

| Operation | Gate | Expression |
|---|---|---|
| **Sum** | XOR | `Sum = A ⊕ B` |
| **Carry** | AND | `Carry = A · B` |

**Half Adder Diagram:**

```text
A ─┬── XOR ── Sum
   │
   └── AND ── Carry
B ─┬── XOR ── Sum (same gate)
   │
   └── AND ── Carry (same gate)
```

> *"Two single bits are added: the XOR gives the sum, and the AND gives the carry."*

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**Phase 4: Simplify by Boolean Law (~15 mins)**

**The Laws of Boolean Algebra:**

| Law | Form 1 (AND) | Form 2 (OR) |
|---|---|---|
| **Annulment** | `A · 0 = 0` | `A + 1 = 1` |
| **Identity** | `A · 1 = A` | `A + 0 = A` |
| **Complement** | `A · Ā = 0` | `A + Ā = 1` |
| **Idempotent** | `A · A = A` | `A + A = A` |
| **Double Negation** | `Ā̄ = A` | |
| **Absorption** | `A · (A + B) = A` | `A + (A · B) = A` |
| **Distributive** | `A · (B + C) = A·B + A·C` | `A + (B·C) = (A+B)·(A+C)` |
| **De Morgan** | `A · B = Ā + B̄` | `A + B = Ā · B̄` |

**Worked Simplification — Named Laws:**

| Step | Expression | Law Used |
|---|---|---|
| 1 | `Z = (A · B̄) + (A · B)` | Original |
| 2 | `Z = A · (B̄ + B)` | Distributive (factor A) |
| 3 | `Z = A · 1` | Complement |
| 4 | `Z = A` | Identity |

**Connecting Back to A1.2.4:**

> *"From the previous lesson,* `(A + B) · (A + C)` *simplifies to* `A + B·C` *by the distributive law—the same result the Karnaugh map gave."*

**Student Task:** Simplify one expression, naming each law, and **verify by re-tabulating** against the original.

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### Part 3: Deliberate Practice — ~20%

**Activity (Pairs):**

| Task | Description |
|---|---|
| Task 1 | Construct a diagram from a given expression; mark against the three criteria |
| Task 2 | Simplify a given expression; name each law and confirm the result by truth table |

**Sentence Frame for Naming Laws:**

> *"By the [law name] law, [expression] simplifies to [result] because ______."*

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### Part 4: Exit Check — ~10%

**Two Items (Readable in Minutes):**

| Question | Purpose |
|---|---|
| **Part A:** "Construct the diagram for `Z = A · (B + C)`" | Checks diagram construction against three criteria |
| **Part B:** "Simplify `Z = A · B + A · B̄`, naming each law" | Checks simplification with named laws |

**Expected Answer (Part B):**

| Step | Expression | Law Used |
|---|---|---|
| 1 | `Z = A · B + A · B̄` | Original |
| 2 | `Z = A · (B + B̄)` | Distributive |
| 3 | `Z = A · 1` | Complement |
| 4 | `Z = A` | Identity |

> **Mark Part A against the three diagram criteria (symbol, connections, labels).**

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## 3. Complete Reference: Seven Gate Symbols

| Gate | Symbol Shape | Rule | Inversion |
|---|---|---|---|
| **AND** | D-shape | 1 only when both inputs are 1 | No |
| **OR** | Curved D-shape | 1 when either input is 1 (including both) | No |
| **NOT** | Triangle with bubble | Flips its single input | Bubble on output |
| **NAND** | AND with bubble | Inverted AND | Bubble on output |
| **NOR** | OR with bubble | Inverted OR | Bubble on output |
| **XOR** | OR with extra curve | 1 only when inputs differ | No |

| Gate | Symbol Shape | Rule | Inversion |
|---|---|---|---|
| **XNOR** | XOR with bubble | 1 only when inputs match | Bubble on output |

### Common Confusions to Address:

| Confusion | Distinction |
|---|---|
| **OR vs. XOR** | OR = 1 if either (including both); XOR = 1 only if exactly one |
| **XOR vs. XNOR** | XNOR is XOR inverted; XNOR = 1 when inputs match |
| **AND vs. NAND** | NAND is AND inverted; NAND = 0 only when both inputs are 1 |

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## 4. Boolean Algebra Laws Summary

| Law | AND Form | OR Form |
|---|---|---|
| **Annulment** | $A \cdot 0 = 0$ | $A + 1 = 1$ |
| **Identity** | $A \cdot 1 = A$ | $A + 0 = A$ |
| **Complement** | $A \cdot \bar{A} = 0$ | $A + \bar{A} = 1$ |
| **Idempotent** | $A \cdot A = A$ | $A + A = A$ |
| **Double Negation** | $\bar{\bar{A}} = A$ | |
| **Absorption** | $A \cdot (A + B) = A$ | $A + (A \cdot B) = A$ |
| **Distributive** | $A \cdot (B + C) = A \cdot B + A \cdot C$ | $A + (B \cdot C) = (A+B) \cdot (A+C)$ |
| **De Morgan** | $A \cdot B = \bar{A} + \bar{B}$ | $A + B = \bar{A} \cdot \bar{B}$ |

**Order of Operations for Boolean Algebra:**

1. Brackets (parentheses) — innermost first
2. NOT ( $\bar{A}$ )
3. AND ( $\cdot$ )
4. OR ( $+$ )

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## 5. Differentiation & Extension

### Support (Removable Scaffolding)

| Support Strategy | Description |
|---|---|
| **Gate Symbol Reference Strip** | Including XNOR; kept on the desk |
| **Fully Worked Simplification** | Each law named; kept as a model to imitate |
| **Sentence Frame** | For naming a law; withdrawn once students name them unprompted |

### Challenge (Deeper and Wider, Not Merely More)

| Challenge Strategy | Description |
|---|---|
| **Apply De Morgan** | Apply De Morgan's theorem to a bracket before simplifying |
| **Count Gates Saved** | Construct a diagram, simplify it, construct the reduced diagram, and count the gates saved |
| **Link to Truth Tables** | Confirm that the original and reduced diagrams are equivalent by truth table (links algebra route back to A1.2.4) |

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## 6. Example Hinge Questions

| Question | Purpose |
|---|---|
| "What are the three core logical operations you know?" | Activates prior knowledge |
| "Draw the XNOR symbol." | Checks recognition of full set of seven gates |
| "Which single change turns an OR into a NOR?" | Checks understanding of the bubble rule |
| "Which gate is 1 only when the inputs match?" | Checks understanding of XNOR |
| "Which law would you use first to simplify `A · B + A · B̄`?" | Checks recognition of factoring opportunity |
| "What is the benefit of simplifying a Boolean expression?" | Checks understanding of purpose (fewer gates) |

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## 7. Teacher Notes: Addressing Student Challenges

| Potential Challenge | Advice/Strategy |
|---|---|
| **Distinguishing OR and XOR** | Emphasise exact phrasing: OR = true if A or B or both; XOR = true if A or B but NOT both. Analogy: "You can have soda OR water (or both)" vs. "You can choose a red shirt XOR a blue shirt (but not both)." |
| **Order of Operations** | Write hierarchy on the board: 1. Brackets, 2. NOT, 3. AND, 4. OR. Students must follow this strictly. |
| **Applying Boolean Laws** | Advise students to look for patterns: Complements (`A + Ā` → 1), Common terms to factor (Distributive), Redundancy (Absorption). |
| **Connecting Algebra to Diagrams** | Continually remind students that every term removed is a physical gate saved. Show "before" (complex) and "after" (simplified) diagram side by side. |

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## 8. Learning Objectives (Summary)

### Knowledge and Understanding

- Identify and state the function of the standard logic gate symbols (AND, OR, NOT, NAND, NOR, XOR, XNOR)
- Apply the laws of Boolean algebra to simplify logical expressions
- Understand how combinations of gates perform complex operations (half adder)

### ATL Skills

| Skill | Description |
|---|---|
| **Self-Management** | Organise multi-step simplification; check own work by re-tabulation |
| **Thinking** | Apply critical thinking to simplify complex expressions |
| **Communication** | Present simplification clearly with named laws |

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## 9. Resources

- **Presentation** (Slide Deck)
- **InThinking eBook** (A1.2.5)
- **Worksheet:** Diagram construction + Boolean simplification
- **Worksheet Answers**

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## Author's Reflection

> "*The main change is that XNOR is now included. The material I started from listed six gates and left it out, so the full set of seven that the syllabus names is taught together. I also gave more of the lesson to students actually drawing circuits, since the command term is Construct and the earlier deck spent most of its time on the gate symbols and the laws rather than on building a diagram.*

> *The laws table and the worked simplifications were already sound, so I kept them and added a set of practice questions, because the laws only stick with use. I added one simple check as well: once students simplify an expression, they build the truth table for their answer and compare it with the original, which tells them whether they are right. The lesson ends by returning to the two circuits from the previous lesson and simplifying one to* `A + B · C` *by algebra, so the truth table, the Karnaugh map, and the laws all reach the same place.*"